Part - 1

Ans. to the que. no. 1

Given,
$$\int_{(0,1)}^{(2,5)} (3x+y)\,dx + (2y-x)\,dy$$

① First of all,

along the straight line $(0,1)$ to $(2,1)$ and

then from $(2,1)$ to $(2,5)$

C_1 $(0,1) \to (2,1)$

$$z = i + (2+i-i)t$$
$$= 2t + i$$

$x = 2t$, $dx = 2dt$

$y = 1$ $dy = 0$

$$\Rightarrow \int_{(0,1)}^{(2,1)} (3x+y)\,dx + (2y-x)\,dy$$

$$= \int_0^1 (6t+1)\,2dt + 0 \qquad [\text{limit } 0 \to 1]$$

$$= \int_0^1 (12t+2)\,dt$$

$$= \left[\frac{12t^2}{2} + 2t \right]_0^1$$

$$= 8$$

<u>C_2 $(2,1) \to (2,5)$</u>

$$z = 2 + i + (2 + 5i - 2 - i)t$$

$$= 2 + i + 4it$$

$$= 2 + i(1 + 4t)$$

$x = 2$, $dx = 0$

$y = 1 + 4t$, $dy = 4dt$

$$\Rightarrow \int_{(2,1)}^{(2,5)} (3x + y)\,dx + (2y - x)\,dy$$

$$= \int_0^1 0 + \left[2(1+4t) - 2\right] 4dt$$ [we know, limit 0 to 1]

$$= \int_0^1 8t \cdot 4dt$$

$$= \int_0^1 32t\,dt$$

$$= \left[\frac{32t^2}{2}\right]_0^1$$

$$= 16$$

$\therefore$ total $= \int_{(0,1)}^{(2,1)} (3x+y)\,dx + (2y-x)\,dy + \int_{(2,1)}^{(2,5)} (3x+y)\,dx + (2y-x)\,dy$

$= 8 + 16$

$= 24$

Ans.

(ii) Along with parabola $y = x^2 + 1$

let,

$x = t \quad , \quad dx = dt$

$y = t^2 + 1 \quad , \quad dy = 2t\,dt$

<u>limit :</u>

x, y	t
$(2,5)$	$t = 2$
$(0,1)$	$t = 0$

Now,

$$\int_{(0,1)}^{(2,5)} (3x+y)\,dx + (2y-x)\,dy$$

$$= \int_0^2 \left[(3t + t^2 + 1)\,dt + \left[2(t^2+1) - t\right] 2t\,dt \right]$$

$$= \int_0^2 (3t + t^2 + 1)\,dt + (4t^3 + 4t - 2t^2)\,dt$$

$$= \int_0^2 (4t^3 + 2t - t^2 + 1)\,dt$$

$$= \left[\frac{4t^4}{4} + \frac{2t^2}{2} - \frac{t^3}{3} + t \right]_0^2 = \frac{88}{3}$$

Ans.

Ans. to the que. no. 2

Given,

$$\int_{|z|=3} \frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-1)(z-2)} dz$$

Now, doing partial fraction,

$$\frac{1}{(z-1)(z-2)} \equiv \frac{A}{(z-1)} + \frac{B}{(z-2)}$$

~~If,~~

$$1 \equiv A(z-2) + B(z-1)$$

If, $z = 1$,

$1 = -A$

$\therefore A = -1$

If $z = 2$,

$1 = B$

$\therefore B = 1$

$$\therefore \frac{1}{(z-1)(z-2)} \equiv \frac{-1}{(z-1)} + \frac{1}{(z-2)}$$

Now,

$$\frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-1)(z-2)} = \frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-2)} - \frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-1)}$$

According to cauchy integral formula

$f(z) = \sin(\pi z^2) + \cos(\pi z^2)$ is analytic inside the region $|z| = 3$

$$\int_{|z|=3} \frac{\sin(\pi z^2)+\cos(\pi z^2)}{z-2}\,dz - \int_{|z|=3} \frac{\sin(\pi z^2)+\cos(\pi z^2)}{z-1}\,dz$$

$$= 2\pi i\left[\sin(\pi.4)+\cos(\pi.4)\right] - 2\pi i\left[\sin(\pi)+\cos(\pi)\right]$$

$$= 2\pi i \,.\, 1 - 2\pi i(-1)$$

$$= 4\pi i$$

Ans.

Ans. to the que. no. 8

Given that,

$$\int_{C:|z|=3} \frac{z}{z^2+4}\,dz$$

~~Therefore~~

$$f(z) = \frac{z}{z^2+4}$$

$$\text{Res}(f, z=2) = \lim_{z\to 2} (z-2) f(z)$$

$$= \lim_{z\to 2} (z-2)\frac{z}{(z-2)(z+2)}$$

z^2-4

$\Rightarrow (z+2)(z-2)$

$\therefore z = 2, -2$

Given,

$|z| = 3$

$|2| = 2 < 3$

$|-2| = 2 < 3$

$$\text{Res}(f, 2) = \lim_{z \to 2} \frac{z}{z+2}$$

$$= \frac{2}{4}$$

$$= \frac{1}{2}$$

for,

$z = -2$

$$\text{Res}(f, -2) = \lim_{z \to -2} (z+2) \frac{z}{(z+2)(z-2)}$$

$$= \lim_{z \to -2} \frac{z}{z-2}$$

$$= \frac{-2}{-4}$$

$$= \frac{1}{2}$$

$\therefore$ sum of residues $= \frac{1}{2} + \frac{1}{2}$

$= 1$

$$\therefore \oint \frac{z}{z^2-4} dz = 2\pi i \times 1$$

$$= 2\pi i$$

Ans,

part -2

Ans. to the que. no.10

Given,

$$f(t) = e^{-2t}\, t\left[\sin(t).\cos(t)\right]$$

$$\mathcal{L}\{\sin(t).\cos(t)\}$$

$$= \frac{1}{2}\mathcal{L}\left\{\sin(2t) + \sin(0)\right\}$$

$$= \frac{1}{2}\mathcal{L}\left\{\sin(2t)\right\}$$

$$= \frac{1}{2}\left[\frac{2}{s^2+4}\right]$$

Now,

$$\mathcal{L}\{t.\sin(t).\cos(t)\}$$

$$= -\frac{d}{ds}\left(\frac{1}{s^2+4}\right)$$

$$= -\frac{-2s}{(s^2+4)^2}$$

$$= \frac{2s}{(s^2+4)^2}$$

Finally,

$$\mathcal{L}\left\{e^{-2t} \cdot t \cdot \sin(t) \cdot \cos(t)\right\}$$

$$= \frac{2(s+2)}{\left[(s+2)^2 + 4\right]^2}$$

$$= \frac{2(s+2)}{(s^2+4s+8)^2}$$

Ans.

Ans. to the que. no. 15

$$\mathcal{L}^{-1}\left\{\frac{6s-4}{s^2-8s-9}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{6s-4}{s^2-2.4.s+4^2-25}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{6s-4}{(s-4)^2-5^2}\right\}$$

$$= e^{4t} \cdot \mathcal{L}^{-1}\left\{\frac{6(s+4)-4}{s^2-5^2}\right\}$$

$$= e^{4t} \cdot \mathcal{L}^{-1}\left\{ \frac{6s+20}{s^2-5^2} \right\}$$

$$= 6 \cdot e^{4t} \mathcal{L}^{-1}\left\{ \frac{s}{s^2-5^2} \right\} + \frac{20}{5} e^{4t} \mathcal{L}^{-1}\left\{ \frac{5}{s^2-5^2} \right\}$$

$$= 6e^{4t} \cosh(5t) + 4e^{4t} \sinh(5t)$$

Ans.

Ans. to the que. no. 16

Given,

$$\frac{s^2+2s+3}{(s^2+2s+2)(s^2+2s+5)} e^{-3\pi s}$$

let,

$$u = s^2+2s$$

$$\frac{u+3}{(u+2)(u+5)} \equiv \frac{A}{u+2} + \frac{B}{u+5}$$

$$(u+3) \equiv A(u+5) + B(u+2)$$

when,

$u = -2$, $1 = 3A$, $A = \frac{1}{3}$

when,

$u = -5$, $-2 = -3B$, $B = \frac{2}{3}$

$$\frac{u+3}{(u+2)(u+5)} = \frac{1/3}{u+2} + \frac{2/3}{u+5}$$

Now,

$$\mathcal{L}^{-1}\left\{\frac{s^2+2s+3}{(s^2+2s+5)(s^2+2s+2)}\right\} = \frac{1}{3}\mathcal{L}^{-1}\left\{\frac{1}{s^2+2s+2}\right\} + \frac{2}{3}\mathcal{L}^{-1}\left\{\frac{1}{s^2+2s+5}\right\}$$

$$= \frac{1}{3}\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2+1^2}\right\} + \frac{2}{3}\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2+2^2}\right\}$$

$$= \frac{1}{3}e^{-t}\mathcal{L}^{-1}\left\{\frac{1}{s^2+1^2}\right\} + \frac{2}{3}e^{-t}\mathcal{L}^{-1}\left\{\frac{2}{s^2+2^2}\right\}$$

$$f(t) = \frac{1}{3}e^{-t}\cdot\sin(t) + \frac{1}{3}e^{-t}\cdot\sin(2t)$$

Finally,

$$\mathcal{L}^{-1}\left\{\frac{s^2+2s+3}{(s^2+2s+5)(s^2+2s+2)}\cdot e^{-3\pi s}\right\}$$

$$= f(t-3\pi)\,u(t-3\pi)$$

$$= \left[\frac{1}{3}e^{-(t-3\pi)}\cdot\sin(t-3\pi) + \frac{2}{3}e^{-(t-3\pi)}\cdot\sin(2(t-3\pi))\right]u(t-3\pi)$$

$$= \left[-\frac{1}{3}e^{(-t+3\pi)}\cdot\sin(t) + \frac{1}{3}e^{(-t+3\pi)}\sin(2t)\right]u(t-3\pi)$$

Ans.